CSE 635: NLP and Text Mining

Spring 2026

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Group Project Description and Requirements

Overview:

The goal of this semester-long project is to provide hands-on experience designing, implementing, evaluating, and demonstrating a complete web mining/ text mining/ social media mining solution based on a combination of natural language processing (NLP), information retrieval (IR) and machine learning (ML) techniques. You are provided a choice of four topics that broadly fall into the **AI for Social Impact** area.We encourage you to use any available online tools or platforms to develop your solution. While there is a quantitative evaluation component in the last stage, you must develop a **live demo system**. This project will satisfy the MS project requirements specified by the CSE department. While the problem definition and evaluation criteria have been fixed, there is ample room for creativity on your part in further enhancement of the solution and implementation.

While we **encourage good prompting** during development, the project will be evaluated based on your **creativity, problem-solving approach, and depth of implementation and evaluation**, rather than reliance on AI-generated solutions.

Be creative, and most importantly, time management is key during the semester. Your project is divided into three phases and more details will be provided before the beginning of each phase:

- **Phase 1**: Submission of project proposal and in-person presentation of your proposal. This includes a comprehensive literature review on your selected topic, a necessary step before you begin designing your system!
- **Phase 2**: Interim report describing the evaluation of the baseline system.
- **Phase 3**: Final submission of technical paper, code, and in-class presentation of your end to end System.

Your project should focus on solving a real problem in a thoughtful, end-to-end way, not just showcasing a model or technique. We want to see how your system gathers information, reasons through it, and delivers something genuinely useful to a user. Make sure you evaluate your work beyond basic correctness by considering usability, reliability, and how it fits within real-world social norms. The system should behave like an agent that can combine information from multiple sources, including external data and references.

You should demonstrate how a real user would interact with your system through a concrete example, and show where personalization adds value. Finally, explain how your system plans its responses and makes decisions, and include some form of LLM-based evaluation or reflection to support your design choices

Note: Only two members are allowed per team.

The project choices are as follows:

- P1: UB Website Chatbot for the CSE Dept
- P2: Multilingual Health-Related Chatbot
- P3: Tutoring/Learning Assistant (Rehabilitation Science)
- P4: Multimodal AAC Chatbot

Project 1 - UB Website Chatbot for the CSE Dept

Background:

The Computer Science and Engineering (CSE) Department at the University at Buffalo hosts extensive information on its website, including details about academic programs, course offerings, credit requirements, faculty, and departmental policies. However, this information is often spread across multiple pages, making it difficult for students to quickly find precise answers to specific questions. This project proposes developing a UB CSE Website Chatbot that serves as a conversational assistant, enabling students and other users to access CSE-related information through natural language queries. The chatbot is designed to answer questions about undergraduate, MS, and PhD programs, course schedules, syllabi, and faculty teaching assignments, while also supporting limited small talk for a natural interaction experience. To ensure reliability and domain relevance, the system incorporates guardrails that restrict responses to CSE-related topics and explicitly notify users when queries fall outside the department's scope.

Dataset:
The dataset for this chatbot should be a custom, structured collection of information extracted from the official University at Buffalo Department of Computer Science and Engineering website (https://engineering.buffalo.edu/computer-science-engineering.html). This site contains sections and hyperlinks that provide detailed information on academics, people, research, and policies.

You can use the following ways to curate the data:

1. **Web Scraping and Crawling**
2. **Manual Curation & Structuring**
3. **Semantic Categorization**

4. **Link Expansion,** etc

Data collection is not strictly limited to the primary URL; any verified external resources directly related to UB CSE, such as official course catalogs or departmental social media are highly encouraged. Please be creative.

Tasks:
To build a high-quality chatbot for the CSE department, the development process is divided into data orchestration, system architecture, and interaction design:

- **Recursive Web Crawling & Content Extraction**: Develop a robust scraper to traverse the UB CSE domain. This includes handling dynamic content and extracting text from academic catalogs, faculty directories, and PDF syllabi.
- **Knowledge Graph Construction**: Beyond simple text chunks, map relationships between entities (e.g., linking a Course ID to its Prerequisites and the Faculty members currently teaching it) to handle complex relational queries.
    - **BONUS:** Based on a query about a course, the bot suggests related research labs or faculty members.
- **Hybrid Retrieval Pipeline**: Implement a system that combines Semantic Search (for general program advice) and Keyword/Exact Match (for specific course codes like "CSE 574") to ensure precision.
    - **BONUS:** Use of re-rankers (both lexical and cross-enc architectures will be rewarded bonus), your UI should show that in logs/ backend panel how chunks are being re-ranked
- **Contextual Guardrail Engineering**: Develop a classification layer to identify out-of-scope prompts (e.g., "Where is the best pizza in Buffalo?") and redirect the user back to CSE-related topics.
- **Dialogue State Management**: Enable the bot to handle follow-up questions (e.g., User: "Who teaches CSE 115?" -> "What are their office hours?") by maintaining a short-term conversation memory.
    - **Requirement:** Personalized memory management will have bonus points: The bot asks the user if they want to personalize their experience, if TRUE, the bot remembers important parts of the conversation to optimize the latency for answering queries.

Evaluation:

- **Faithfulness & Grounding:** Use the **RAGAS framework** to ensure the chatbot's answers are derived strictly from the scraped UB CSE data, minimizing "hallucinations" about degree requirements.
- **Retrieval Precision (Hit Rate):** Measure how often the top-k retrieved documents actually contain the answer to a student's query.
- **Task Success Rate (Human-in-the-loop):** Live evaluation in class
- **Latency Benchmarking:** Ensure the response time (Time to First Token) remains under **2 seconds** to maintain a natural conversational flow.

- **Robustness Testing:** Test the "Guardrails" by asking non-CSE or provocative questions to ensure the bot maintains its departmental persona.

Bonus Points for implementation **of one or more** of the above features highlighted in green.

References:

1. https://engineering.buffalo.edu/computer-science-engineering.html
2. https://engineering.buffalo.edu/computer-science-engineering/research/research-areas/artificial-intelligence/artificial-intelligence-and-machine-learning-and-data-mining.html
3. https://engineering.buffalo.edu/computer-science-engineering/graduate.html
4. https://engineering.buffalo.edu/computer-science-engineering/graduate/degrees-and-programs/ms-in-computer-science-and-engineering.html
5. https://www.buffalo.edu/ubnow/Driven-to-discover.host.html/content/shared/university/news/ub-reporter-articles/podcast/Srihari-better-chatbots.detail.html
6. https://www.instagram.com/ub_cse/?hl=en

Project 2 - Conversational AI Chatbot for Multilingual Mental Health Screening

Background

Traditional mental health screening in clinical settings often relies on the direct administration of standardized questionnaires like the PHQ-9 (depression), GAD-7 (anxiety), or CAGE-AID (substance use). It uses rigid "yes/no" questionnaires that can feel like a clinical test, often leading to "survey fatigue" or guarded answers.

The objective of this project is to build a multilingual, voice-capable conversational agent that acts as a "screening partner". Instead of asking "Have you felt bad about yourself?", the agent engages in a natural, empathetic dialogue to infer these clinical constructs from the user's narrative. This approach aims to minimize patient burden while maintaining high diagnostic accuracy through adaptive questioning strategies

Dataset

- Input Data: Multilingual transcripts and audio recordings **in any 2 languages, which is English and one other from these - Hindi, Kannada, or Tamil.**

- Patient Profiles: Basic socio-demographic information (age, occupation, etc.).

- Gold Standard Labels: Item-level scores (0–3) for validated scales assigned by expert psychiatrists to specific symptoms (items) based on official medical checklists.

  To get the "Gold Standard" labels, you can use **public clinical datasets** (like DAIC-WOZ Distress Analysis Interview Corpus (DAIC)) that already feature expert-assigned scores. Alternatively, you can **grade transcripts yourselves** by matching patient symptoms against the official DSM-5-TR (Diagnostic and Statistical Manual of Mental Disorders) medical criteria.

- Knowledge Base: Clinical guidelines, such as the *DSM-5-TR*, used to ground the bot in medical definitions rather than general intuition.

Tasks:

Please note that the model/architecture details given here are only for guidance, you are free to use any model/architecture of your choice to complete the tasks.

- **Task 1: Smart Screening & Confidence-Based Scoring**
  - **The Goal:** Build a "Smart Map" for the screening process. Instead of a robotic, fixed checklist (Question 1, then Question 2), the bot decides the next best question based on the user's previous answers. Have an initial chit-chat conversation to ease the user into the conversation.
  - **The "Graph" Architecture:** Think of the screening as a **flexible map**. If a user mentions they aren't sleeping well, the bot can "jump" to a specific section about sleep habits. If they seem fine in one area, the bot moves quickly; if they mention something concerning, it circles back to learn more.
  - **Confidence Scoring (The "Trust" Meter):** For every screening category (like "Anxiety Levels" or "Sleep Quality"), the bot calculates a **Confidence Score**.
    - **High Confidence:** The bot has enough information to understand the user's state in that category and moves to the next topic.
    - **Low Confidence:** The bot feels it doesn't have enough "evidence" yet. It triggers a **"Follow-up Branch"** to ask more gentle, probing questions until it is sure it has an accurate picture.
- **Task 2: Conversational Inference Engine (LLM-based)**
  - Goal: Analyze open-ended chat to assign clinical scores (0–3) for specific questionnaire items.
  - You can use models like Llama 3 or GPT-4o. Implement a Chain of Thought (CoT) approach where the model first extracts "evidence snippets" from the dialogue before assigning a final score.
- **Task 3: Safety Trigger System**
  - Goal: Detect immediate risks such as thoughts of self-harm or emotional escalation.
  - You can use high-priority classifiers like SBERT (Bi-Encoders) to flag high-risk semantic patterns for immediate human escalation.

Evaluation & Validation

- Disclosure Efficiency: Measuring the minimum amount of conversational evidence (number of "turns") required to reach a stable score.
- Safety Accuracy: Evaluating the bot's ability to correctly identify and stop for high-risk emotional escalations.
- Latency: Measures the time from the student's input to the bot's response, ensuring the "Hybrid Architecture" remains fast enough for a real-time conversation.
- Discourse Effectiveness: Evaluates how well the bot stayed on track to cover all the necessary questionnaire points without leaving out important information..

Bonus:

You can implement one or more of the below features for bonus points.

- Gamification: Using interactive "nudges" to encourage users to share longer, more descriptive narratives.
- Linguistic Personalization: Mirroring the user's style, pacing, or verbosity to improve disclosure confidence.

References:

Enhancing Natural Language Processing in Multilingual Chatbots: Explores improving cross-cultural communication through advanced NLP in multilingual bots.

Diagnostic Evaluation and AI-Inferred Scores: Discusses the correlation between AI models and standardized clinical assessments.

Conversational Agents for Screening: Research on building natural dialogue systems for clinical data collection.

Digital Personal Data Protection (DPDP 2023): Official guidelines for handling sensitive patient data in the Indian context.

Clinical Validation of Screening Tools: A study focusing on the efficacy and validation of mental health screening constructs.

Project 3 - Socratic-OT Multimodal AI Tutor (Tutoring/Learning Assistant - Rehabilitation Science)

Background:

Occupational Therapy (OT) is a healthcare profession focused on helping people achieve independence in daily life. To succeed, students must master Gross Anatomy and Neuroscience. These are "bottleneck" courses due to their high content density and heavy reliance on complex 3D visual relationships.

The Problem: Students lack 1:1 support outside of class. Standard AI (like ChatGPT) often gives answers away, preventing students from building the "clinical reasoning" needed for the NBCOT certification exam. The Solution: You will build a Socratic AI Tutor that uses Retrieval-Augmented Generation (RAG) to guide students using only faculty-approved textbooks and diagrams.

What is "Socratic" Tutoring?

In this project, "Socratic" means the bot follows a "Tutor-not-Teller" philosophy.

- The Goal: Lead the student to the answer by asking questions based on the textbook context.
- The Constraint: The bot is strictly forbidden from providing a direct definition or answer in the first two turns. It must use the retrieved data to ask a "leading" question that triggers the student's memory.

Dataset: Input Sources

Gather and structure data into a "Knowledge Base."

- Textual Data (PDF/HTML): Use open-source medical texts.
  - *Source:* OpenStax Anatomy & Physiology 2e
- Visual Data (Images): Curate 20–30 anatomical diagrams with labels.
  - *Source:* AnatomyTOOL or MedPix®
- Format: Text should be stored in a Vector Database (e.g., ChromaDB, FAISS). Images must have associated metadata (JSON) describing the structures shown.

Tasks:

Please note that the model/architecture details given here are only for guidance, you are free to use any model/architecture of your choice to complete the tasks.

Task 1: Content Retrieval & Knowledge Masking

- **Mechanism:** Develop a RAG pipeline to retrieve sections from OT textbooks (e.g., neuroscience/anatomy).
- **The "Mask":** The agent identifies the target answer but "masks" it from the student.
- **Logic:** The LLM generates progressive hints based on the hidden data without revealing the specific term, forcing student discovery.

Task 1: Adaptive Conversation Strategy

- **Phase 1 (Rapport):** Open with context-aware chitchat (e.g., "Ready for the lab exam?") to build engagement.
- **Phase 2 (Tutoring):** Pivot to the masked content using a Socratic loop.
- **Agentic Control:** A **Manager Agent** monitors the dialogue state, ensuring the transition from social to pedagogical is seamless.

Task 3: Synthesis & Assessment

- **Evaluation:** Move beyond multiple-choice. The agent prompts the student to apply the concept to a clinical scenario (e.g., "Explain how this nerve lesion impacts the body").
- **Outcome:** The agent compares the student's reasoning against the gold-standard textbook data and provides a mastery summary.

Task 4: Multimodal Diagram Tutoring:

- Use a Vision-Language Model (VLM) like GPT-4o or LLaVA. When a student uploads an image and asks, "What is this muscle?"The bot must identify it and ask the student about its function or insertion point. Use a combination of high-reasoning API calls (OpenAI/Claude) and local Open Source LLMs (Llama 3/Mistral) to manage costs and data privacy.

Task 5: Interactive Memory:

- Implement a session-based memory where the bot remembers past student mistakes and proactively revisits those topics. (tutoring modules- about 15mins per session)

Evaluation Metrics:

- Groundedness (Quantitative): Use the RAGAS framework to measure *Faithfulness*. If the bot provides info not found in the provided textbook, it fails the "hallucination" test.

- Socratic Purity (Qualitative): You must provide 5 conversation transcripts. The bot must not "leak" the final answer before the 3rd turn.

- Multimodal Performance: The bot must correctly identify anatomical structures in a "blind test" set of diagrams.
- Generalizability: Demonstrate that your code can work for a different subject (e.g., Physics) simply by swapping the Vector Database.

Bonus:

You can implement one or more of the below features for bonus points.

- Personalization: A dashboard showing the student's "weak spots" based on chat history.
- Accessibility: Integrating "Text-to-Speech" or "Speech-to-Text" to support diverse learning needs.

References:

- MS Articulate 360: Linear, static branching. Why is your AI more flexible?
- **Boodlebox**: A general-purpose tool. How does your OT-specific grounding make it safer for healthcare?
- **Harvard Learning Lab (Tutor-not-Teller):** Research their "System 2 Thinking" AI experiments which force models to reason before responding.
- **Khanmigo (Khan Academy):** The industry benchmark for AI tutoring. Study their technical blogs on using "System Prompts" to prevent direct answers.
- **RAGAS Framework:** The standard for evaluating Retrieval-Augmented Generation. Use this to measure **Faithfulness** (no hallucinations) and **Answer Relevance**.
    - *Resource:* RAGAS Documentation
- **SocraticLM Paper:** Review "SocraticLM: Exploring Socratic Personalized Teaching with LLMs" (NeurIPS) for strategies on fine-tuning LLMs for guided questioning.
- **Multimodal VLM Research:** Explore documentation for **GPT-4o (Vision)** or **LLaVA** to understand how to prompt models for spatial anatomical reasoning.

Project 4- Multimodal AAC Chatbot

Background:

This project evolves the integration of Large Language Models (LLMs) with Augmentative and Alternative Communication (AAC) by moving away from static fine-tuning toward a training-free, agentic orchestration model. Traditional systems often result in "voice appropriation," where the AI speaks for the user rather than as the user. To solve this, the system uses an agentic layer to dynamically decompose conversational intent and retrieve personal context in real-time. A major focus of this update is multimodal synchronization, enabling the chatbot to incorporate non-verbal cues such as facial gestures and "air signing" to ensure the generated response aligns with the user's physical expression and emotional state.

Dataset:

The system utilizes a non-parametric approach to personal data, requiring the following curated resources, please create synthetic data using LLMs (this will be later used for evaluation as well):

- **Structured Profile Data**: A machine-readable schema summarizing communication traits, access needs, stylistic preferences (e.g., "short and humorous"), and personal background.
- **Heterogeneous Personal Memory**: A collection of unstructured data including autobiographical narratives, social media posts, and past communication logs.
- **Multimodal Signal Data**: Streams or logs of facial affect and spatial "air writing" movements to be used as intent-boosters for response generation
    - **Hand "Like" Symbol**: If the user provides a "thumbs up" or similar hand gesture, the agentic controller immediately boosts the ranking of affirmative candidate responses and increases the "happy" tone multiplier.
    - **Head Nod (Dissatisfaction)**: Contrary to generic interpretations, the system can be calibrated to recognize specific head movements (e.g., a sharp nod or tilt) as markers of disagreement or dissatisfaction, triggering the "Turnaround Option" for clarification.
    - **[BONUS] Gaze-Based Retrieval Activation**: Sustained eye gaze on specific interface regions or external objects can act as a "focus trigger," signaling the retrieval module to prioritize semantic buckets related to the observed object (e.g., gazing at a family photo triggers the "Family" memory bucket).
    - **[BONUS] Conflict Resolution**: The system should monitor vocalizations that occur simultaneously with "air signing" to resolve ambiguities in intent, using the combined signal to refine the plan execution before final response generation.

Tasks:

- **Agentic Intent Decomposition**: Implement a controller that breaks partner queries into sub-queries classified as Personal, Contextual, or Open-domain to route them to the correct retrieval pools.

- **Multimodal Input Mapping**: Develop a sensing layer to translate facial gestures and "air signing" into stylistic constraints for the LLM (e.g., a smile-triggering a more positive lexical choice).
- **Contextual Semantic Bucketing**: Group personal data chunks into thematic "buckets" (e.g., family, daily routine, medical) using clustering techniques to enable fast, topic-aware retrieval.
    - **BONUS**: Go beyond basic retrieval by implementing "Bucket Priors". This system should adjust retrieval weights based on the current conversational topic, favoring entire semantic buckets (e.g., "Medical" or "Friends") that have consistently yielded accepted responses in the current session.
- **Real-Time Retrieval & Reranking**: Use dense semantic embeddings to fetch relevant memories and apply reranking to prioritize evidence that matches the current conversation.
    - **BONUS**: Latency-Optimized Fallback Strategies: Implement a tiered reasoning system that monitors response time. If a high-fidelity generation or external web search exceeds a specific latency threshold (e.g., 5 seconds), the system should automatically switch to a faster, lightweight local model to maintain conversational flow.
- **Training-Free Response Generation**: Fuse retrieved evidence with the structured data into an augmented prompt, allowing the model to adapt its persona without explicit fine-tuning.
    - **BONUS**: Implement a mechanism where a user's selection of a candidate response triggers an immediate, lightweight update to the retrieval index.

Evaluation & Validation:

- **Factual Faithfulness:** Use NLI-based metrics to measure Groundedness (statements entailed by evidence) and Hallucination rates.
- **Communication Efficiency**: Benchmark Response Latency (target < 6 seconds) to ensure the system supports natural conversational flow.
- **Perceived Authenticity**: Human evaluation by experts and partners on a 5-point Likert scale to assess if the bot reflects the user's unique expressive style. Live in class evaluation
- **Multimodal Alignment**: Accuracy of the system in reflecting non-verbal physical inputs within the generated text output.

Bonus Points for implementation **of one or more** of the above features highlighted in green.

References: Will be shared in class

Grading Schema:

Please note that as the respective milestone approaches, more details will be given on the exact date and submission details on Piazza.

1. **Milestone 1 (10%):** Project Proposal (week of March 3)

- Literature Review
- Project objectives
- Dataset features to be implemented

- Solution architecture

- Evaluation methodology
- Computational resources plan
- Project Plan
- Presentation of the project plan

2. **Milestone 2 (10%):** Baseline results (week of April 3)

3. **Milestone 3 (25%):** Final Project Report and Presentation (May 6th)

Submission Details:

a. Submission of the report must follow the technical paper format of ACL (Within 5 pages) . The report must be a comprehensive analysis of your project, backed with graphs and visualizations of your observations.
Please refrain from just stating the meaning of an image or a visualization , instead provide your inference/observation about it.

b. For each phase, the working code and related documents(in .pdf and .pptx(for the presentation) format) must be submitted in a .zip file titled
ProjectPhaseNumber_UBid1_UBid2.zip.(Eg: ProjectPhase1_jdoe_rain.zip)